EXPERIMENT 10

Applying Performance Optimization Techniques (Indexing, Subforms, and Query Tuning) to a Zoho Creator Application

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Application: Vehicle Service & Repair Request Management

Aim

To study and apply performance-oriented record management techniques in the Vehicle Service & Repair Request Management application in Zoho Creator, using report filters, Quick Filters, sorting/criteria concepts, and data-validation settings.

Introduction

Performance optimization helps users locate relevant records with less unnecessary browsing and supports reliable data management. In this experiment, report filtering and Quick Filters were configured for service requests. Indexing and subforms are included as optimization concepts from the experiment topic; the screenshots provided do not verify that a manual database index or a new subform was created.

Procedure

Step 1: Indexing concept and record searching

• Identified Vehicle_Registration_Number as a useful field for finding a vehicle's service request.
• Examined report filtering using a registration-number criterion.
• Studied the role of indexing in improving retrieval speed. The screenshots show a report criterion, not a confirmed manual database-index configuration.

Step 2: Subforms

• Reviewed the purpose of a subform for storing multiple related service-detail rows under one service request.
• A subform named Additional_Service_Details with Service_Name, Service_Cost, and Service_Notes was proposed as a possible extension. It is not recorded as implemented because no saved subform configuration or submitted subform record was provided.

Step 3: Query tuning using report filters

• Opened the Service_Request_Report settings.
• Configured a report criterion using Vehicle_Registration_Number and a sample value.
• Retained Quick Filters for Service_Type and Preferred_Service_Date so users can narrow displayed records.
• Reviewed the live report to verify that service-request records could be displayed and inspected.

Observations

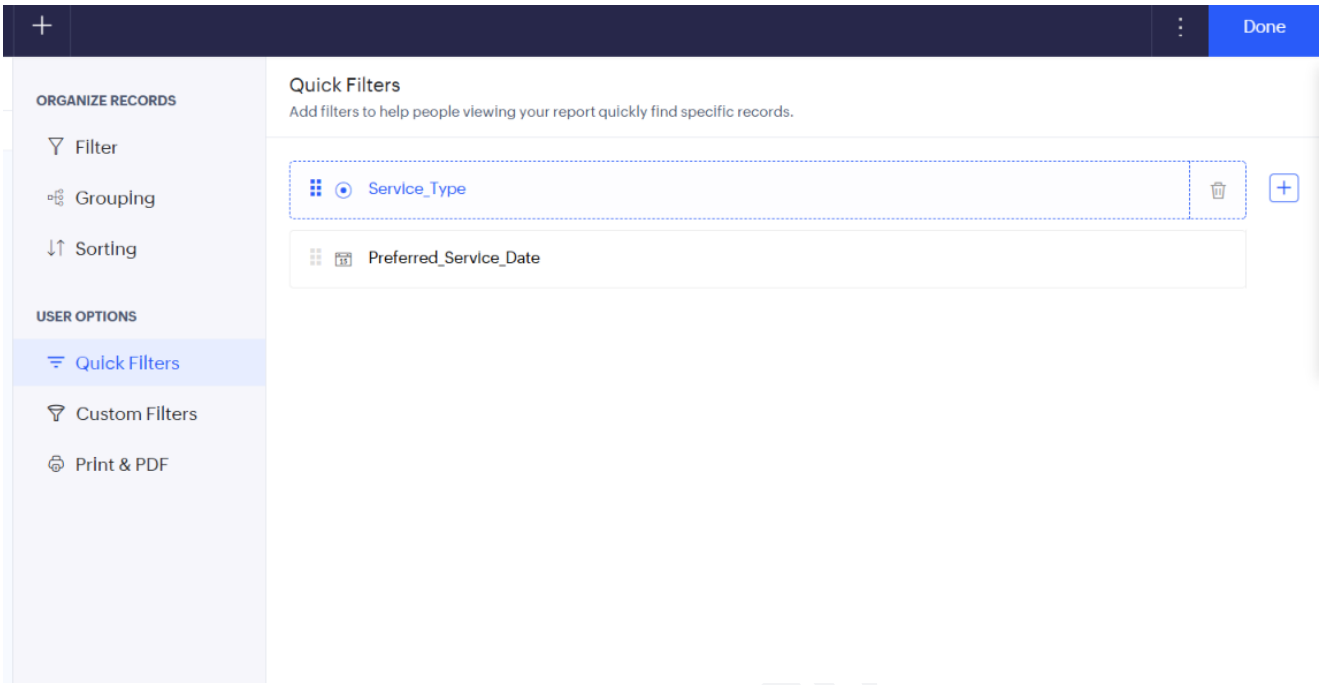
1. Quick Filters were configured for Service_Type and Preferred_Service_Date.
2. A report filter criterion was set for Vehicle_Registration_Number.
3. The live report displayed service-request records, including a filtered view and a multi-record view.
4. The screenshots support report filtering and record retrieval. They do not independently prove manual indexing or subform implementation.

Result

Report filtering and Quick Filters were configured and reviewed in the Vehicle Service & Repair Request Management application. The exercise demonstrated how criteria-based retrieval can help users locate relevant service requests. Indexing and subforms were documented as optimization concepts, while implementation of a manual index or new subform remains unverified by the supplied evidence.

Screenshot Evidence 1

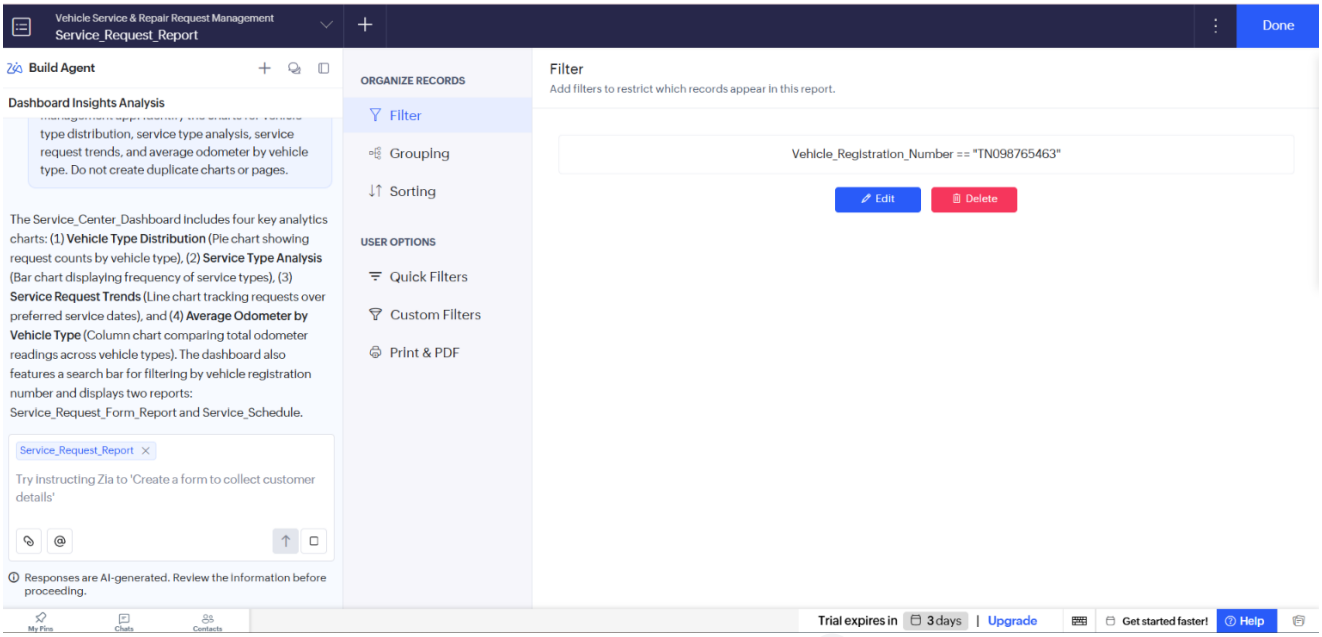
Figure 1: Quick Filters configured for Service_Type and Preferred_Service_Date.



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Screenshot Evidence 2

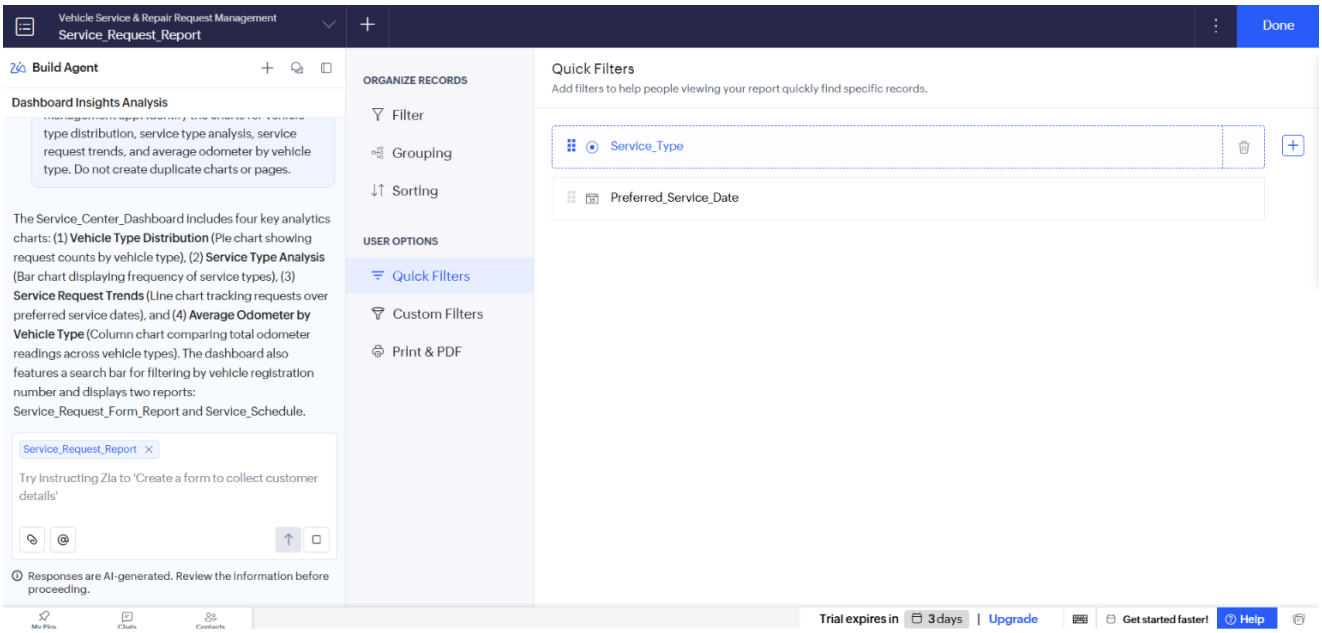
Figure 2: Report filter criterion using Vehicle_Registration_Number.



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Screenshot Evidence 3

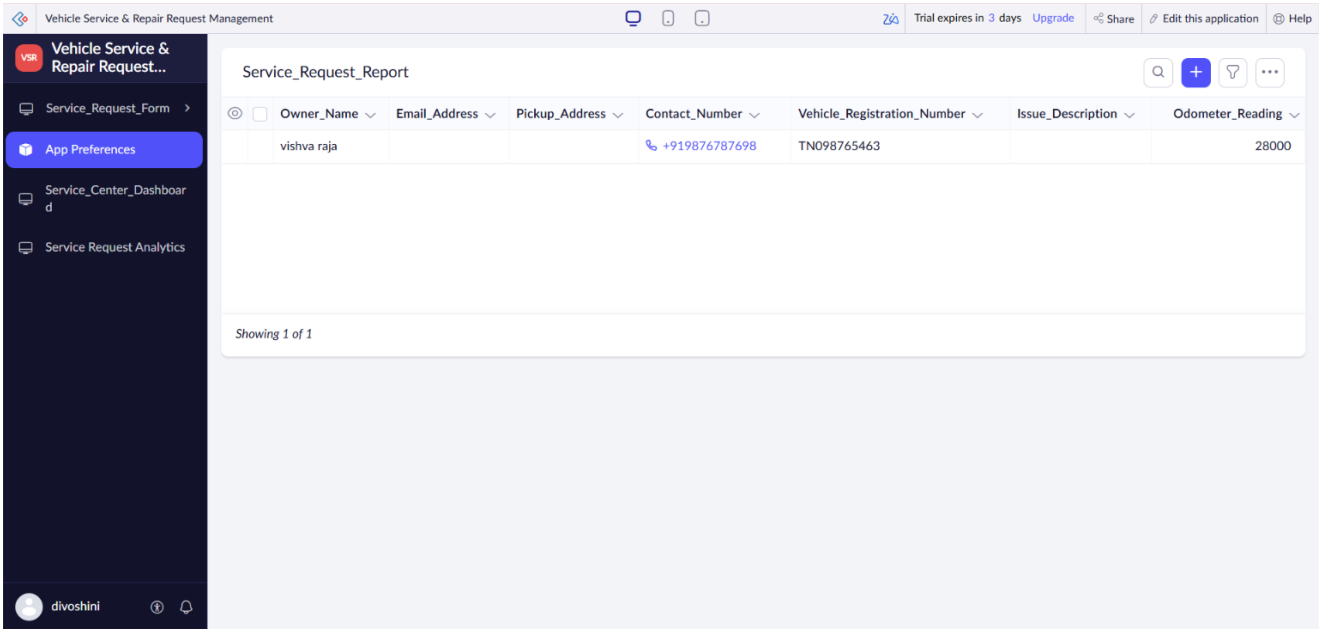
Figure 3: Quick Filters settings in the report builder.



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Screenshot Evidence 4

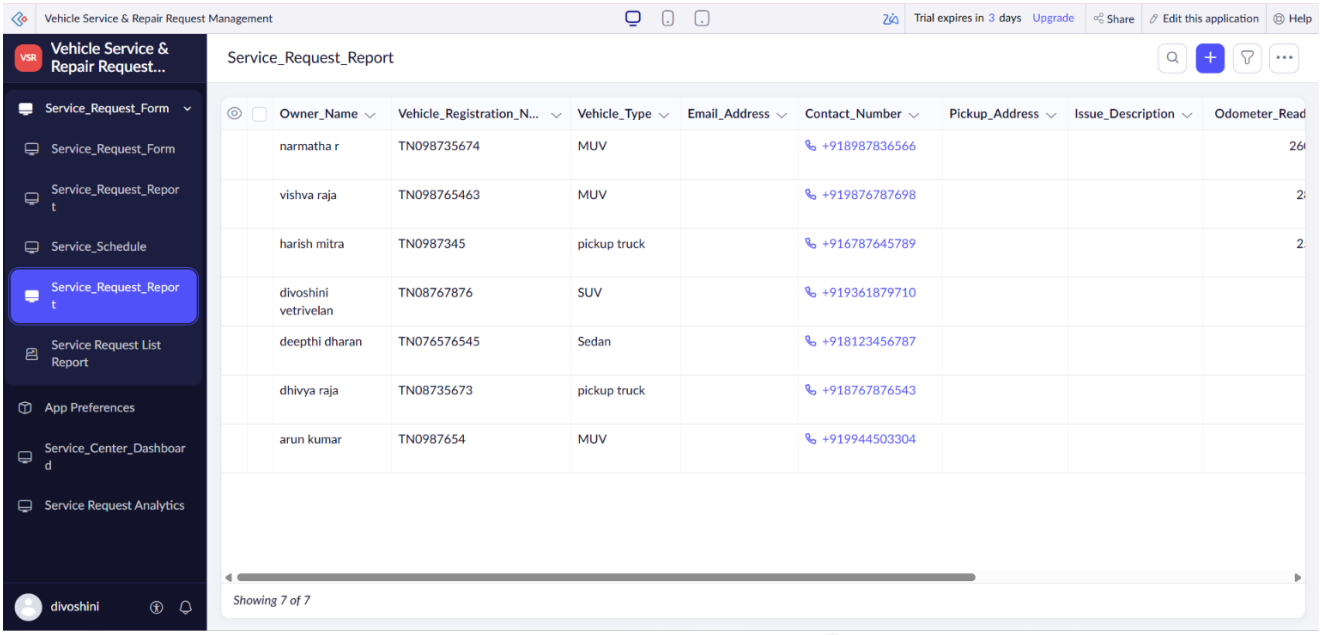
Figure 4: Service request report showing a matching record.



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Screenshot Evidence 5

Figure 5: Service request report displaying multiple records for review.



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